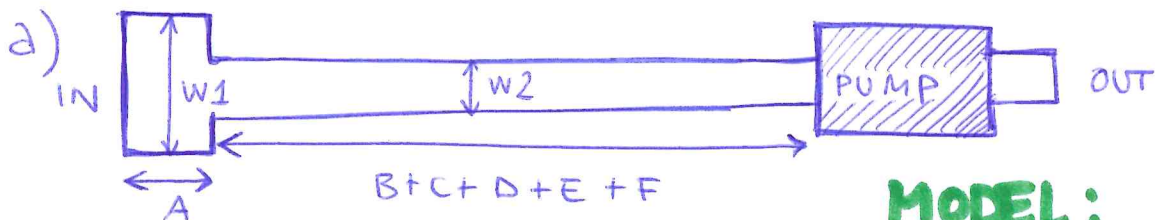
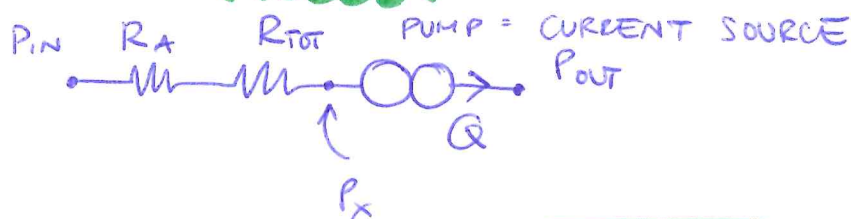


EX. 1

$$P_{IN} = P_{OUT} = P_0 \quad 1/4$$



MODEL:



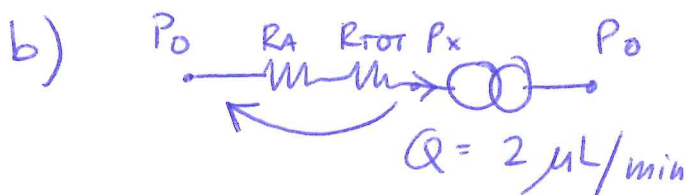
$$R_A = \frac{12 \cdot \eta \cdot L}{(1 - 0.63 \frac{h}{w}) h^3 \cdot w} = \frac{12 \cdot \eta \cdot L}{h^3 \cdot w}$$

$R_A |_{AIR} = 1.4 \cdot 10^{11}$
 $R_A |_{WATER} = 8 \cdot 10^{12} \left[\frac{Pa \cdot s}{m^3} \right]$

$$R_{TOT} = \frac{8 \cdot \eta \cdot L}{\pi r_{eq}^4}$$

$r_{eq} = 10 \mu m$
 $R_{TOT} |_{AIR} = 7.3 \cdot 10^{13}$
 $R_{TOT} |_{WATER} = 4 \cdot 10^{15} \left[\frac{Pa \cdot s}{m^3} \right]$

$$L = B + C + D + E + F = 15 \text{ mm} \quad \text{DOMINANT}$$



$$P_x = P_0 - (R_A + R_{TOT}) \cdot Q$$

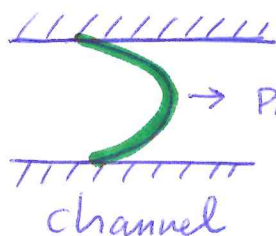
$$P_x = P_0 - \frac{2 \cdot 10^{-6}}{1000} \frac{m^3}{60 \cdot s} \cdot 4 \cdot 10^{15} \frac{Pa \cdot s}{m^3} = 101 \text{ kPa} - 133 \text{ kPa}$$

$$= -32 \text{ kPa}$$

$$V = \frac{Q}{h \cdot w}$$

$V_A = \frac{2 \mu L / min}{20 \mu m \cdot 200 \mu m} = 8.3 \text{ mm/s}$
 $V_{TOT} = \frac{2 \mu L / min}{20 \mu m \cdot 20 \mu m} = 83.3 \text{ mm/s}$

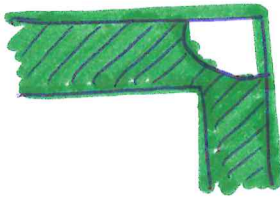
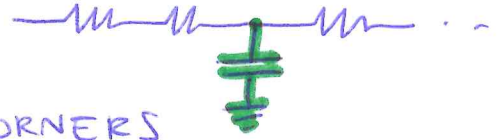
VERY FAST!



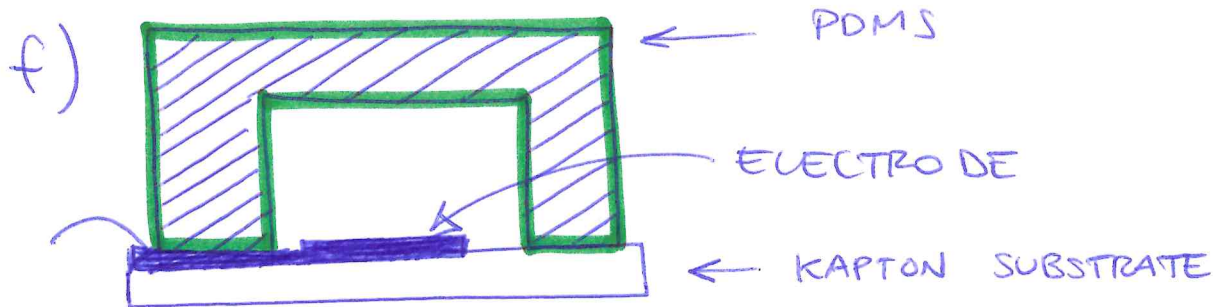
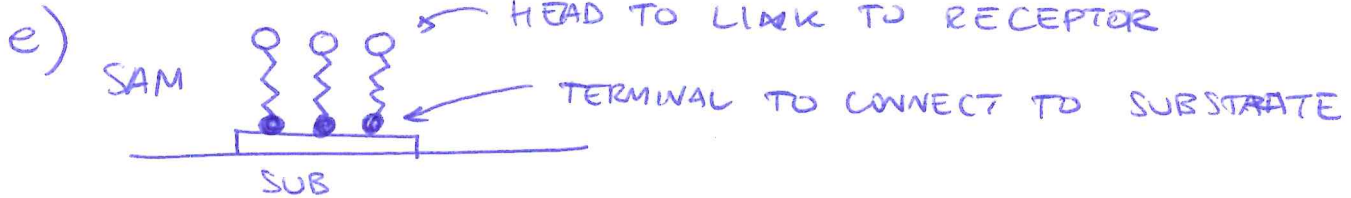
→ PARABOLIC VELOCITY DISTRIBUTION

channel

c)

BUBBLE $\rightarrow$ FLUIDIC CAPACITANCE

MOST LIKELY TO STOP IN THE CORNERS

d) $\gamma = \gamma(T)$ if $T \uparrow \Rightarrow \gamma \downarrow \Rightarrow R \downarrow$ 

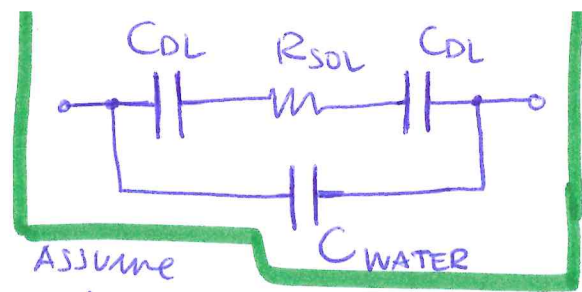
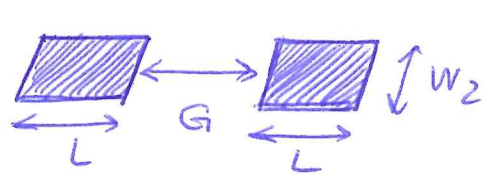
- COPPER: LOW-COST BUT REACTIVE!

- GOLD: EXPENSIVE, NON STANDARD, BUT ELECTROCHEMICALLY STABLE.

g) DIGITAL MICROFLUIDICS, e.g. ELECTROWETTING

EX. 2

a)



MODUL 7

5/4

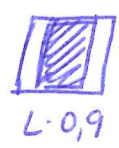
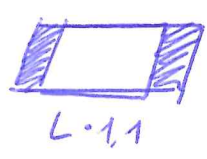
$R_{SOL} \leftarrow$ CONFORMAL MAPPING, ASSUME $10K\Omega$
 $= \left[\rho \cdot \frac{1}{\kappa} \right]$

$$C_{DL} = \frac{0,1 \text{ pF}}{\mu m^2} \cdot L \cdot W_2 = \frac{0,1 \text{ pF}}{\mu m^2} \cdot 60 \mu m \cdot 20 \mu m = 120 \text{ pF}$$

$$C_W = \epsilon_0 \cdot \epsilon_{WATER} \cdot \underset{8}{K} \quad K \text{ from CONFORMAL MAPPING}$$

$\rightarrow$ USUALLY NEGLIGIBLE BELOW 350 MHz for PBS

$L(1,1)$



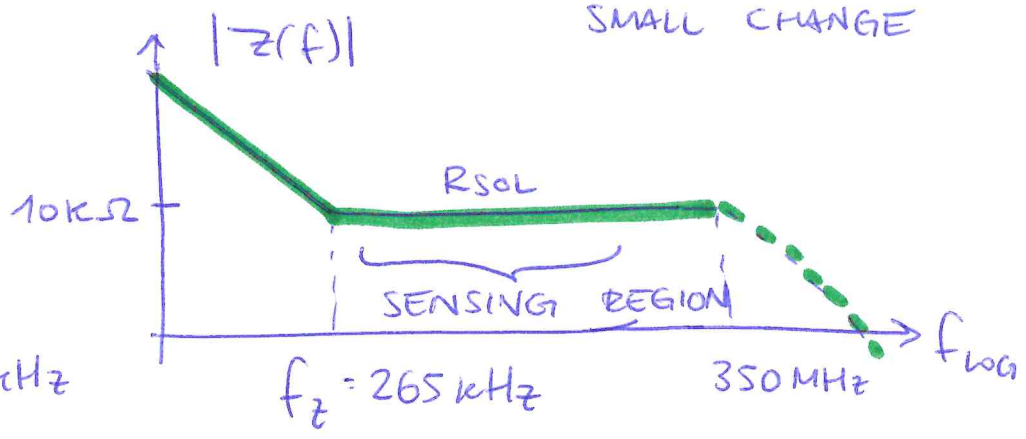
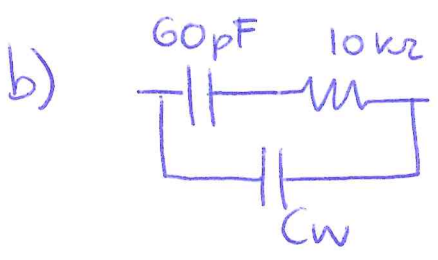
$$C_{DL1} = C_{DL} \cdot 0,9 = 108 \text{ pF}$$

$$C_{DL2} = C_{DL} \cdot 1,1 = 132 \text{ pF}$$

$$C_{DL} |_{SERIE} |_{10} = 60 \text{ pF}$$

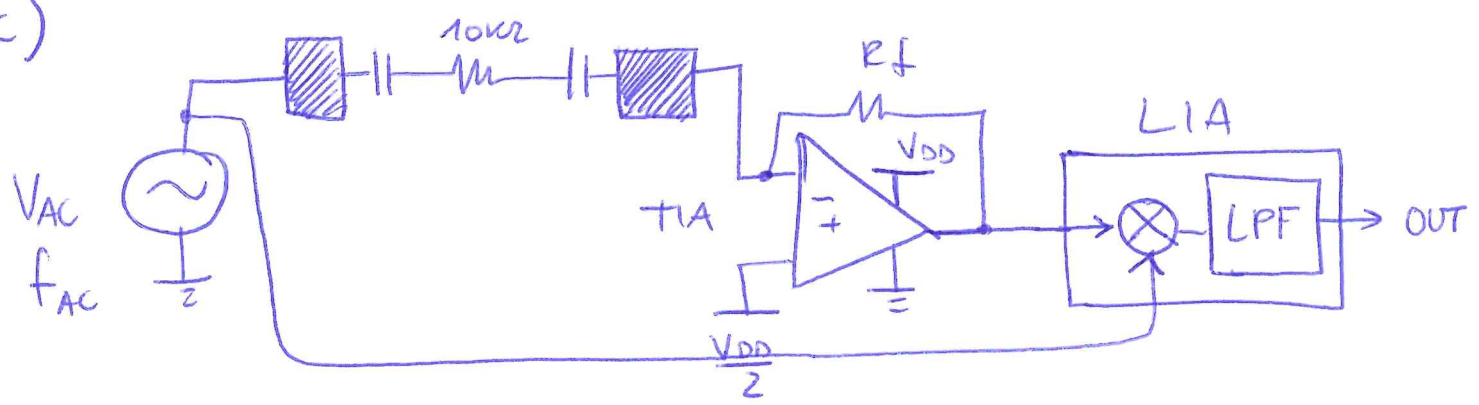
$$C_{DL} = \frac{108 \cdot 132}{108 + 132} = 59,4 \text{ pF}$$

SMALL CHANGE

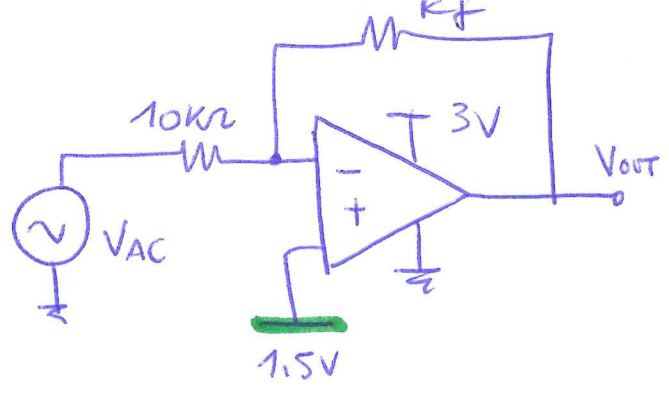


$$f_z = \frac{1}{2\pi C_{DL} \cdot R} = 265 \text{ kHz}$$

c)



$$f_{AC} > f_z \rightarrow f_{AC} = 2 \text{ MHz}$$



I_{set}
 $V_{AC} = 100 \mu V$

$V_{OUT} = V_{AC} \cdot \left| \frac{R_f}{R_{in}} \right| = 1.5V$

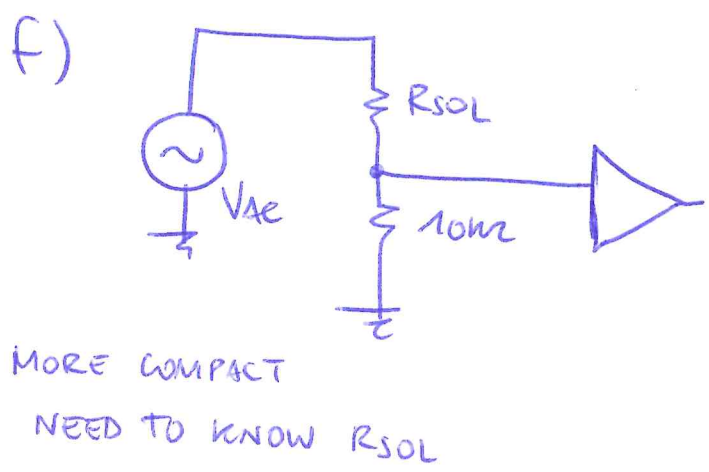
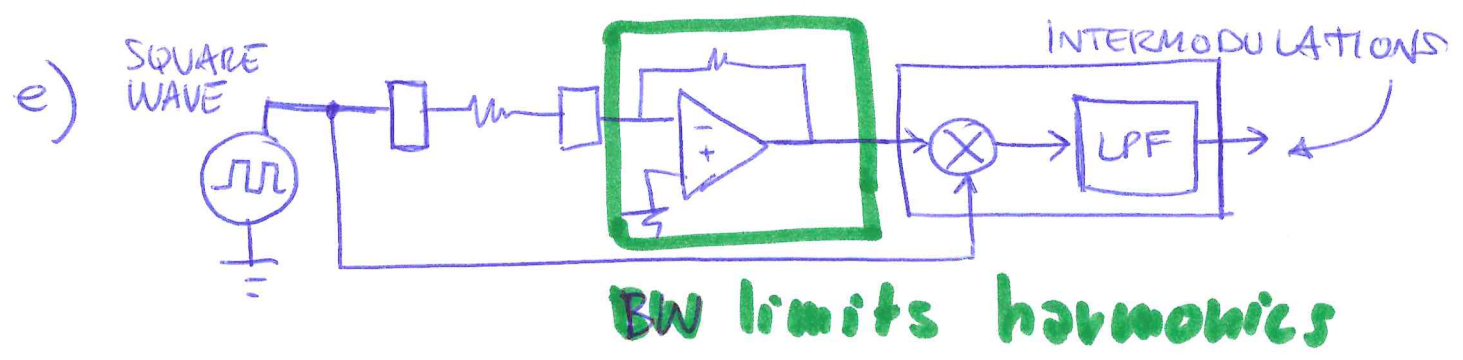
$R_f = R_{sol} \cdot \frac{V_{OUT}}{V_{AC}} = 10k\Omega \cdot \frac{1500 \mu V}{100 \mu V} = 150k\Omega$

Check BW: $BW = \frac{1}{2\pi \cdot 150k\Omega \cdot 0.2pF} = 5.3 MHz > 2MHz$
 (OK, TIGHT)

$LIA/BW \sim \frac{1}{1\mu s} = 1kHz$ COULD TAKE SOME MARGIN $\sim 10kHz$

d) $\sigma_i = \sqrt{\frac{4kT}{R_{sol}} \cdot \frac{\pi}{2} \cdot BW} = 1nA_{rms}$
 (Note: R_{sol} is labeled as $10\mu\Omega$ in the original image)

$BW = (1nA)^2 \cdot \frac{2}{\pi} \cdot \frac{R_{sol}}{4 \cdot \frac{kT}{q}} = 382.6 kHz$
 $C_F = 41.6 pF$
 (Note: $\frac{kT}{q}$ is labeled as $26mV$ in the original image)



- g) CMOS
- | | |
|--------------------------------|------------------------|
| <u>PRO</u> | <u>CON</u> |
| - MINIMIZE DISTANCE (STUMP) | - SUBSTRATE PARASITICS |
| - MINIATURIZED MASS PRODUCTION | - COST... |
| | - MATERIALS |
| | - PACKAGE |